

1. 数制和编码

(1) 进制转换

① $N = (a_{n-1} a_{n-2} \dots a_1 a_0 . a_{-1} a_{-2} \dots a_{-m})_r$

$= \left(\sum_{k=-m}^{n-1} a_k r^k \right)_{10}$

② 十进制 $\rightarrow$ k进制

整数 除以k取余, 逆序排列

小数 乘k取整, 顺序排列

$(173)_{10} = (110101101)_2$

$173 = 2 \times 86 + 1$
 $86 = 2 \times 43 + 0$
 $43 = 2 \times 21 + 1$
 $21 = 2 \times 10 + 1$
 $10 = 2 \times 5 + 0$
 $5 = 2 \times 2 + 1$
 $2 = 2 \times 1 + 0$
 $1 = 2 \times 0 + 1$ 终止

$(173.875)_{10} = (10101101.1101)_2$

$0.875 \times 2 = 1.75 = 1 + 0.75$
 $0.75 \times 2 = 1.5 = 1 + 0.5$
 $0.5 \times 2 = 1.0 = 1 + 0.0$
 $0.1000 \times 2 = 1.0000 = 1 + 0.0000$ 终止

(2) 有符号数的表示

$\begin{cases} +: 0 \\ -: r-1 \end{cases} \rightarrow \text{原码, 补码}$

① 补运算: $[Nr] = r^n - Nr$, n为位数

② 补码: $A = a_{n-1} a_{n-2} \dots a_0 = a_{n-1} \cdot (-r^{n-1}) + \sum_{k=0}^{n-2} a_k r^k$
 $\in [-2^{n-1}, 2^{n-1}-1]$

(2) 编码

① BCD 码

加法: BCD $\leftrightarrow$ binary

② 格雷码

构造: 翻转 $\rightarrow$ 填 0/1

② 定理

1 幂等性: $a+a=a, a \cdot a=a$

$a+a = (a+a) \cdot 1 = (a+a)(a+a') = a+(a \cdot a')$
 $= a+0 = a$

$a \cdot a = (a \cdot a) + 0 = a \cdot a + a \cdot a' = a(a+a')$
 $= a \cdot 1 = a$

2 空单元: $a \cdot 0 = 0, a+1=1$

$a \cdot 0 = a \cdot (a \cdot a') = (a \cdot a) \cdot a' = a \cdot a' = 0$

$a+1 = a+(a+a') = (a+a)+a' = a+a' = 1$

3 自反律: $(a')' = a$

$\begin{cases} a'(a')' = 0 \\ a'+(a')' = 1 \end{cases} \begin{cases} a'a=0 \\ a'+a=0 \end{cases} \Rightarrow (a')' = a$

4 吸收律: $a+ab=a, a \cdot (a+b)=a$

$a+ab = a \cdot 1 + ab = a(1+b) = a(b+1) = a \cdot 1 = a$

$a \cdot (a+b) = (a+0) \cdot (a+b) = a+(0 \cdot b)$

$= a+(b \cdot 0) = a+0 = a$

5 $X+X'Y=X+Y, X \cdot (X'+Y)=XY$

$X+X'Y = (X+X')(X+Y) = 1(X+Y) = X+Y$

$X(X'+Y) = XX'+XY = XY$ 对, 有分配律!

6 德摩根律

7 对偶原理

8 $(X+Y)(X'+Z) = XZ + X'Y$ 对偶

$XY + X'Z = (X+Z)(X'+Y)$

$(X+Y)(X'+Z) = XZ + X'Y + YZ$

$= XZ + X'Y + YZX + YZX'$

$= XZ + X'Y$

9 $XY + YZ + X'Z = XZ + X'Y$

$(X+Y)(Y+Z)(X'+Z) = (X+Y)(X'+Z)$

$XY + YZ + X'Z = XY + XYZ + X'YZ + X'Z$

$= XY + X'Z$

10 香农扩展定理

$f(x_1, x_2, \dots, x_n) = x_1 \cdot f(1, x_2, \dots, x_n) + \bar{x}_1 \cdot f(0, x_2, \dots, x_n)$
 $= [x_1 + f(0, x_2, \dots, x_n)] [\bar{x}_1 + f(0, x_2, \dots, x_n)]$

3. 布尔函数 (开关函数)

① 真值表

② 积之和 (与或式) $\rightarrow$ 最小项范式 (析取范式)

③ 和之积 (或与式) $\rightarrow$ 最大项范式 (合取范式)

2. 布尔代数

① 公理

公理 1 封闭性: $a \cdot b \in B, a+b \in B$

2 交换律: $a \cdot b = b \cdot a, a+b = b+a$

3 结合律: $(a \cdot b) \cdot c = a \cdot (b \cdot c), (a+b)+c = a+(b+c)$

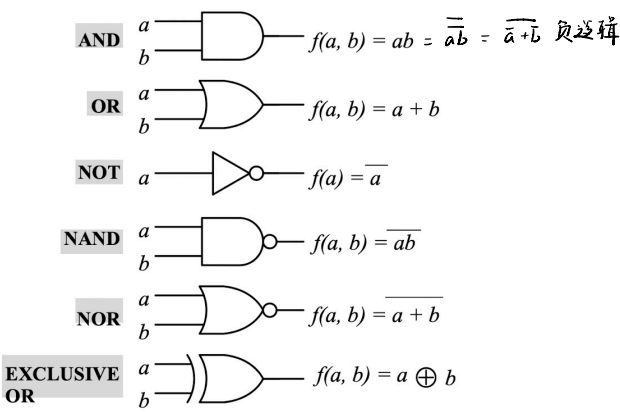
4 恒等性: $a+0=a, a \cdot 1=a$

5 分配律: $(a+b) \cdot c = a \cdot c + b \cdot c, (a \cdot b) + c = a \cdot c + b \cdot c$

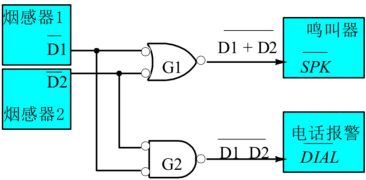
6 互补性: $a+a'=1, a \cdot a'=0$

1. 逻辑门和电路

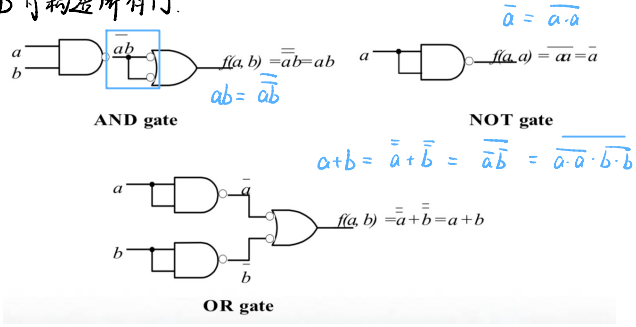
① 电信号与逻辑值：正逻辑 / 负逻辑



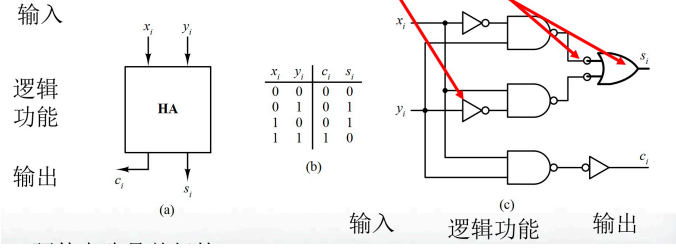
$$\begin{cases} SPK = D_1 + D_2 \\ DIAL = D_1 D_2 \end{cases} \Rightarrow \begin{cases} SPK_{out} = \overline{SPK} = \overline{D_1 + D_2} = \overline{D_1} \cdot \overline{D_2} \\ DIAL_{out} = \overline{DIAL} = \overline{D_1 D_2} = \overline{D_1} + \overline{D_2} \end{cases}$$



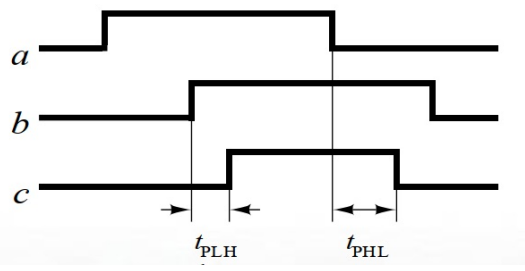
NAND 可构造所有门。



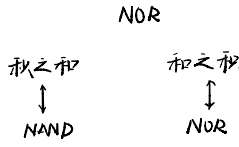
$$s_i(x_i, y_i) = \overline{x_i} \overline{y_i} + x_i y_i$$
$$c_i(x_i, y_i) = x_i y_i$$



② 传播延迟: 一般 $t_{PHL} > t_{PLH}$

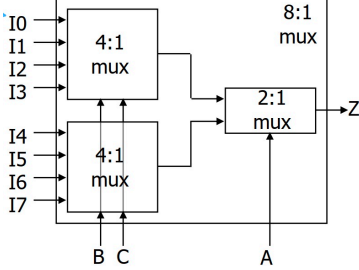


1. AND-OR 变换到 NAND 电路: 圈泡移动法



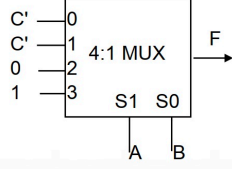
2. 多路选择器

(1) $2^n:1$ MUX 实现任意 n 变量布尔函数



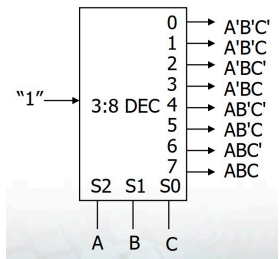
(2) $2^{n-1}:1$ MUX 实现任意 n 变量布尔函数

A	B	C	F
0	0	0	1 C'
0	0	1	0
0	1	0	1 C'
0	1	1	0
1	0	0	0 0
1	0	1	0
1	1	0	1 1
1	1	1	1



3. 多路分配器 / 译码器

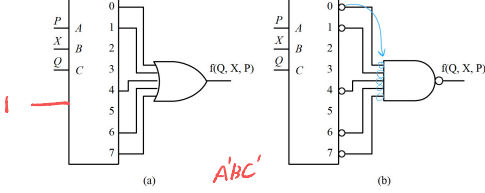
(1) $n:2^n$ decoder 实现任意 n 变量布尔函数



“最小项生成器”
输出加非即可生成最大项

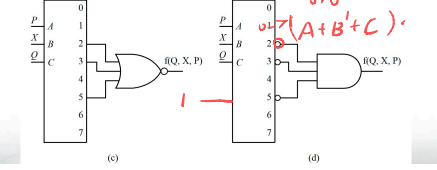
(a) $f(Q, X, P) = m_0 + m_1 + m_4 + m_6 + m_7$

(b) $f(Q, X, P) = \overline{m_0 \cdot m_1 \cdot m_4 \cdot m_6 \cdot m_7}$



(c) $f(Q, X, P) = m_2 + m_3 + m_5$

(d) $f(Q, X, P) = \overline{m_2 \cdot m_3 \cdot m_5} = M_2 \cdot M_3 \cdot M_5$



4. 卡诺图

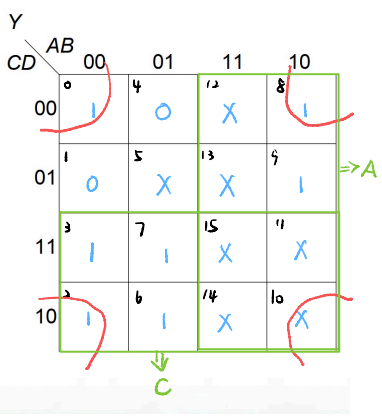
(1) 化简布尔表达式

目标：组合最小项/最大项

原则：用最小的方形覆盖所有 0/1

- 1. 计算每个最小项的相邻度 *注意边用!*
- 2. 选择未被覆盖的相邻度最小的最小项，如果有多个相同的选项，任选一个。 *贪心*
- 3. 生成包含该最小项的尽可能大的方形组，如果存在多个相同规模的组，任选一个。
- 4. 重复步骤2和步骤3，直到所有的最小项都被覆盖。

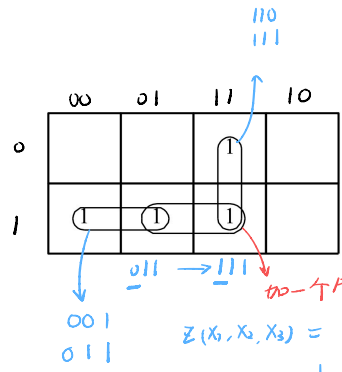
A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	X
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X



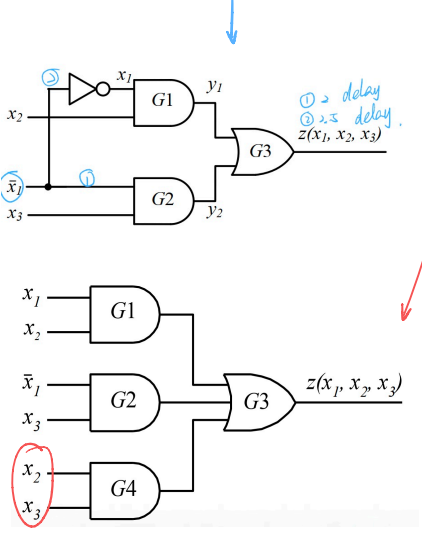
0000
1000 $\Rightarrow \bar{B}\bar{D}$
0010
1010
 $\bar{B}\bar{D} + A + C$

(2) 处理静态冒险

1 冒险:
0 冒险:



a. 一个方组对应一个门
b. 相邻且不在同一组的 1/0 导致 1/0 冒险。



总结

